A

Major Project

On

QUANTUMHEALTHCHAIN:A POST-QUANTUM KYBER- FALCON KEY EXCHANGE AND BLOCKCHAIN-BACKED FRAMEWORK FOR TAMPERPROOF ELECTRONIC HEALTH RECORDS

(Submitted in partial fulfilment of the requirements for the award of Degree)

BACHELOR OF TECHNOLOGY

In

INFORMATION TECHNOLOGY

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CERTIFICATE

This is to certify that the project entitled **"QUANTUMHEALTHCHAIN : A POST-QUANTUM KYBER-FALCON KEY EXCHANGE AND BLOCKCHAIN-BACKED FRAMEWORK FOR TAMPERPROOF ELECTRONIC HEALTH RECORDS" being submitted by A. Sahithi (227R1A1265), K. Raghunath (227R1A1298) & B. Nikhil (227R1A1271)** in partial fulfilment of the requirements for the award of the degree of B. Tech in Information Technology to the Jawaharlal Nehru Technological University Hyderabad, during the year 2025-26.

The results embodied in this thesis have not been submitted to any other University or Institute for the award of any degree or diploma.

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ABSTRACT

Quantum Resistance Blockchain for Secure Health Data aims to address the growing vulnerabilities in healthcare systems due to outdated encryption and centralized storage. In India, healthcare data breaches have steadily increased, with incidents like the 2021 breach at a government health database affecting over 100 million records. The rising digitalization of health records, telemedicine, and e-governance demands quantum-resistant cryptographic systems to safeguard patient privacy. As the country moves towards digital health ecosystems such as Ayushman Bharat Digital Mission (ABDM), integrating blockchain with post-quantum security becomes vital to ensure tamper-proof, decentralized, and future-proof health data management. The objective is to build a secure, decentralized, and future-ready healthcare data system by integrating quantum-resistant cryptography with blockchain, ensuring data confidentiality, immutability, and machine learning- assisted diagnosis in 30 words. Manual systems for healthcare data management involve physical records, spreadsheets, or simple relational databases where patient records are entered and shared manually between departments, leading to inefficiencies, delays, and lack of centralized access control. Drawbacks of manual systems include the risk of human error, paper deterioration, data duplication, unauthorized access, difficulty in maintaining history, no real-time analysis, and a complete lack of transparency or audit trails for file handling. The motivation for this research stems from the limitations of manual and semi-digital systems that fail to offer scalable security. By integrating blockchain and post-quantum cryptography, the system overcomes threats posed by quantum attacks, providing immutable logs, secure communication, and privacy-preserving analytics in a tamper-proof architecture. The proposed system employs machine learning models to analyse encrypted health files after decryption, assisting in diagnosis or risk prediction. The entire process is secured with post-quantum cryptographic schemes like FALCON for signing and KYBER for key encapsulation. Blockchain is used to record all file upload, download, and access actions, offering traceability and tamper-resistance.

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1.INTRODUCTION

1. INTRODUCTION

The digital transformation of healthcare requires secure and intelligent systems to manage patient data. Traditional storage and communication methods are not very secure and can lead to data breaches. This project proposes a secure healthcare system that combines Post- Quantum Cryptography (KYBER and FALCON) for strong encryption, Blockchain for secure and transparent record keeping, and Machine Learning for analyzing medical data. The aim is to create a complete system to upload, encrypt, share, analyze, and retrieve healthcare data securely.

Recently, cyberattacks on hospitals have increased. Hackers may steal patient data or launch ransomware attacks, where they lock hospital data and demand money to unlock it. These attacks can happen due to outdated systems or malicious email links. Security problems also occur when patient records are transferred between hospitals. Attackers may send malware, steal patient data during transfer, or disrupt communication. To solve these issues, the proposed system uses blockchain technology to store and verify medical information securely.

The system also allows secure sharing of patient records between hospitals using encrypted session keys. In addition, insurance companies can be verified and registered on the blockchain, so patient insurance details can be accessed securely without repeatedly submitting documents.

1.1 PROJECT PURPOSE

The primary purpose of this project is to develop a secure and intelligent healthcare data management system that protects sensitive patient information from cyberattacks and unauthorized access. In many hospitals, medical records are stored in centralized systems, which makes them vulnerable to data breaches, hacking, and data manipulation. Therefore, this project aims to create a more secure and reliable platform for storing, sharing, and analyzing healthcare data.

The system uses Post-Quantum Cryptography algorithms such as KYBER and FALCON to encrypt patient files and protect them from both current and future threats, including attacks from quantum computers. It also uses Blockchain technology to record all data transactions in a decentralized and tamper-proof ledger, ensuring transparency, trust, and secure access control.

In addition, the project integrates Machine Learning models to analyze decrypted medical data and provide predictive insights that can help doctors in diagnosis and decision-making. The system also allows secure communication and collaboration between doctors, patients, and healthcare institutions, ensuring that only authorized users can access the medical information.

Overall, the main purpose of this project is to create a future-ready healthcare system that ensures data security, privacy, transparency, and intelligent analysis of medical records while enabling safe and efficient sharing of healthcare information.

1.2 PROJECT FEATURES

This project provides several important features to ensure secure and intelligent healthcare data management.

Secure File Upload and Storage: The system allows patients, doctors, and hospitals to securely upload and store medical records such as prescriptions, reports, and diagnostic results. All uploaded files are protected within the system so that unauthorized users cannot access or modify the data. This ensures that sensitive healthcare information remains safe and confidential.

Post-Quantum Encryption: The project uses post-quantum cryptography algorithms such as KYBER and FALCON to protect patient data. These algorithms encrypt the medical files before storing or sharing them. Post-quantum encryption is designed to resist attacks from both classical computers and future quantum computers, making the system more secure than traditional encryption methods.

Blockchain-Based Logging: All data transactions in the system are recorded on a blockchain network. Blockchain acts as a decentralized and tamper-proof ledger, meaning once information is recorded, it cannot be altered or deleted. This feature ensures data integrity, transparency, and traceability of every action performed in the system.

Secure File Sharing: The platform enables secure sharing of medical records between hospitals, doctors, and patients. When files are transferred, they remain encrypted and can only be accessed by authorized users with proper permissions. This prevents attackers from intercepting or stealing sensitive healthcare data during transmission.

Access Control and Authentication: The system provides role-based access control, where different users such as doctors, patients, and administrators have different permissions. Authentication mechanisms ensure that only verified users can access the system, thereby protecting patient privacy and preventing unauthorized data access.

Data Integrity and Privacy Protection: By combining encryption and blockchain, the system ensures that patient data remains accurate, secure, and private. Any attempt to change or manipulate the stored information can be detected easily, which helps maintain trust in the healthcare system.

2.LITERATURE SURVEY

2. LITERATURE SURVEY

Ryu et al. [1] proposed a new authentication protocol to prevent and address such issues within hospital systems. The proposed protocol encrypts medical records on a private blockchain, allowing them to be securely shared among institutions, hospitals, and insurance companies, ensuring data recovery even if a ransomware attack paralyzes the server. Additionally, the protocol facilitates the systematic sharing of patient medical records between hospitals or between hospitals and insurance companies by distributing session keys. In this study, They demonstrate that the proposed protocol provides 11 security properties, including forward and backward secrecy, user untraceability, and resistance to replay attacks. They also evaluate the communication and computational costs, proving that the protocol is feasible for practical use.

Zhao et al. [2] implemented a decentralized management system with enhanced security and operational efficiency. Considering the scenario of EMRs requiring confirmation of multiple doctors to improve safety, the proposed solution leverages Shamir's Secret Sharing to enable multi-party authorization, thereby enhancing privacy protection. Meanwhile, the scheme utilizes Bloom filter and vector operation to achieve efficient data search. The proposed method maintains rigorous EMR protection while improving the search efficiency of EMRs. Experimental results demonstrate that, compared to existing methodologies, the proposed scheme enhances security during EMR sharing processes. It achieves higher efficiency in index generation and trapdoor generation while reducing keyword search time. This scheme provides reliable technical support for the development of intelligent healthcare systems.

Wenhua et al. [3] mentioned advantages and disadvantages motivated us to provide an advancement and comprehensive study regarding the applicability of blockchain technology. They focuses on the healthcare security issues in blockchain and sorts out the security risks in six layers of blockchain technology by comparing and analyzing existing security measures. It also explores and defines the different security attacks and challenges when applying blockchain technology, which promotes theoretical and robust security protocol development in the current and future distributed work environment.

Alsadhan et al. [4] introduced significant privacy concerns, particularly regarding data privacy. IoMT devices often collect and store large amounts of data about patients' health. These data could be used to track patients' movements, monitor their health habits, and even predict their future health risks. This extensive data collection and surveillance could be a major invasion of patient privacy. Thus, privacy- preserving research in an IoMT context is an important area of research that aims to mitigate these privacy issues. They comprehensively applies the PRISMA methodology to analyze, review, classify, and compare current approaches of preserving patient data privacy within IoMT blockchain-based healthcare environments.

Sharma et al. [5] employed to protect these transactions against such attacks. However, the security of blockchain relies on the computational complexity of certain mathematical functions, and because of the evolution of quantum computers, its security may be breached in real-time in the near future. Therefore, researchers are focusing on combining quantum key distribution (QKD) with blockchain to enhance blockchain network security. This new technology is known as quantum-secured blockchain. This article describes different attacks in optical networks and provides a solution to protect networks against security attacks by employing quantum-secured blockchain in optical networks. It provides a brief overview of blockchain technology with its security loopholes, and focuses on QKD, which makes blockchain technology more robust against quantum attacks. Next, the article provides a broad view of quantum-secured blockchain technology.

Ghosh et al. [6] evaluated 144 articles that discuss the importance and limits of using blockchain technologies to improve healthcare operations. The objective is to demonstrate the technology's potential uses and highlight the difficulties and possible sectors for future blockchain research in the healthcare domain. The paper starts with an extensive background study of blockchain and its features. Then, the paper focuses on providing an extensive literature review of the selected articles to highlight the current research themes in blockchain-based healthcare systems. After that, major application areas along with the solutions provided by blockchain in healthcare systems are pointed out.

Liu et al. [7] developed a quantum-secure blockchain (QSB) scheme by introducing a consensus mechanism—quantum proof of authority (QPoA) signature (IQS)—wherein QPoA is used for new block generation and IQS is used for transaction signing and verification. Second, QPoA is developed by adopting a quantum voting protocol to achieve secure and efficient decentralization for the blockchain system, and a quantum random number generator (QRNG) is deployed for randomized leader node election to protect the blockchain system from centralized attacks like distributed denial of service (DDoS). Compared to previous work, our scheme is more practical and efficient without sacrificing security, greatly contributing to better addressing the challenges in the quantum era. Overall, our scheme presents a feasible solution for blockchain systems against quantum computing attacks through a quantum strategy, contributing toward quantum-secured blockchain in the quantum era.

Ajakwe et al. [8] investigated the mainstream security challenges associated with the transmissibility of medical records, the evolutionary security strategies for maintaining confidentiality, and the existential enablers of trustworthy and transparent authorization and authentication before data transmission can be carried out. The review adopted the PRSIMA-SPIDER methodology for a systematic review of 122 articles, comprising 9 surveys (7.37%) for qualitative analysis, 109 technical papers (89.34%), and 4 online reports (3.27%) for quantitative studies. The review outcome indicates that the sensitivity and confidentiality of a highly classified document, such as a medical record, demand unabridged authorization by the owner, unquestionable preservation by the host, untainted transparency in transmission, unbiased traceability, and ubiquitous security, which blockchain technology guarantees, although at the infancy stage.

Venugopal et al. [9] combined the privacy and security advantages of encryption and decentralized networks offered by Blockchain technology with the scalability of distributed systems to provide an effective secure platform for big data applications. The framework is based on the principles of confidentiality and immutability to ensure the security and privacy of health care data.

Alabdulatif et al. [10] proposed methodology addresses computational efficiency and scalability issues via the implementation of lightweight cryptographic techniques, achieving an average authentication delay of 0.059 milliseconds, which represents a 4000-fold improvement compared to current approaches. The model exhibits a significant decrease in

memory use, requiring just 0.0198 MB in contrast to the 96.98 MB required by benchmark models, and attains an average signature verification duration of 0.00092 milliseconds. The findings demonstrate the model's capability for safe, efficient, and scalable applications in e-health, which guarantees privacy and adherence to regulatory norms.

Enaya et al. [11] explored blockchain applications in the IoT, focusing on security, automation, scalability, and data sharing. Industry-specific applications, including supply chain management, smart cities, and healthcare, highlight the potential of blockchains to optimize operations, ensure compliance, and foster innovation. Additionally, blockchain technology enables robust audit trails, enhances accountability, and reduces fraud in sensitive IoT applications, such as finance and healthcare. The synergy between blockchains and the IoT creates a secure and transparent platform for managing device interoperability and data exchange, fostering seamless communication between diverse IoT components. Furthermore, this paper discusses layer 2 scaling techniques and tokenization to address scalability, ownership, monetization, and cost challenges, providing practical solutions for real-world deployments. Future directions emphasize integrating blockchain systems with artificial intelligence (AI), machine learning (ML), and edge computing, offering groundbreaking capabilities to further revolutionize IoT ecosystems.

Ohize et al. [12] presented a survey of the latest trends in the development of e- voting systems, focusing on the integration of blockchain technology as a promising solution to address various concerns in e-voting, including security, transparency, auditability, and voting integrity. They survey is important because existing survey articles do not cover the latest advancements in blockchain technology for e-voting, particularly as it relates to architecture, global trends, and current concerns in the development.

Thus, we address this gap by providing an encompassing overview of architectures, developments, concerns, and solutions in e-voting systems based on the use of blockchain technology. Specifically, a concise summary of the information necessary for implementing blockchain-based e-voting solutions is provided. Furthermore, we discuss recent advances in blockchain systems, which aim to enhance scalability and performance in large-scale voting scenarios. We also highlight the fact that the implementation of blockchain-based e-voting systems faces challenges, including cybersecurity risks, resource intensity, and the need for robust infrastructure, which must be addressed to ensure the scalability and reliability of these systems.

Taherdoost et al. [13] processed by utilizing blockchain technology. By systematically examining articles published from 2017 to 2022, this review addresses the existing gap by methodically discussing the state, research trends, and challenges of blockchain in medical data exchange. The number of articles on this issue has increased, reflecting the growing importance and interest in blockchain research for medical data exchange. Recent blockchain-based medical data sharing advances include safe healthcare management systems, health data architectures, smart contract frameworks, and encryption approaches. The evaluation examines medical data encryption, blockchain networks, and how the Internet of Things (IoT) improves hospital workflows. The findings show that blockchain can improve patient care and healthcare services by securely sharing data.

Li et al. [14] provided a secure execution environment for SQLCipher, isolating all sensitive operations of healthcare data from the untrusted environment to ensure the confidentiality and integrity of the data. Additionally, we design a TEE-empowered session key generation protocol that enables secure authentication and key sharing for both parties involved in data sharing. Finally, we implement TrustHealth using Hyperledger Fabric and ARM TrustZone. Through security and performance evaluation, TrustHealth is shown to securely process massive encrypted data flows at a rate of 5000 records per second, affirming the feasibility of our proposed scheme. We believe that TrustHealth offers valuable guidelines for the design and implementation of similar systems, and ensuring the privacy and security of eHealth systems.

2.1 REVIEW OF RELATED WORK

1. Blockchain-Based Healthcare Systems

Many researchers have explored the use of blockchain technology to improve the security and transparency of healthcare systems. Blockchain provides a decentralized and tamper-proof ledger where medical records can be stored and shared securely. Studies show that blockchain can help maintain data integrity, traceability, and secure access control while allowing hospitals and insurance companies to exchange patient information safely. It also provides an audit trail that records every transaction, helping organizations monitor access and detect unauthorized activities.

2. Secure Authentication and Data Sharing

Several studies focus on developing secure authentication protocols and data sharing mechanisms in healthcare environments. These systems ensure that only authorized users such as doctors, hospitals, and insurance providers can access patient records. Researchers have proposed protocols that use encrypted communication and session keys to protect data during transmission. Multi-party authorization techniques are also used in some systems, where multiple healthcare professionals must approve access to sensitive medical records, thereby improving privacy and preventing unauthorized modifications.

3. Privacy Protection in IoT and IoMT Healthcare Systems

With the rapid growth of Internet of Medical Things (IoMT) devices such as wearable health sensors and smart medical equipment, healthcare systems now collect large volumes of patient health data in real time. However, this also introduces privacy and security risks, as sensitive patient information may be exposed during data transmission or storage. Researchers have proposed privacy-preserving mechanisms that combine encryption, secure communication protocols, and blockchain technology to ensure that patient data remains confidential while enabling remote monitoring and digital healthcare services.

4. Integration of Blockchain with Artificial Intelligence

Recent research has focused on integrating blockchain with artificial intelligence and machine learning technologies. In these systems, blockchain ensures secure and transparent storage of healthcare data, while machine learning models analyze the data to support medical diagnosis, disease prediction, and treatment planning. This integration enables healthcare providers to gain valuable insights from medical records while maintaining strong security and trust in the data management process.

5. Post-Quantum Cryptography for Healthcare Security

Traditional encryption algorithms such as RSA and ECC are widely used in healthcare systems to protect sensitive data. However, the advancement of quantum computing may make
these algorithms vulnerable in the future. To address this challenge, researchers are exploring post-quantum cryptography algorithms that are resistant to quantum attacks. These algorithms provide stronger security for long-term protection of healthcare data and are considered essential for building future-ready healthcare security systems.

2.2 DEFINITION OF PROBLEM STATEMENT

The healthcare sector is highly vulnerable to data breaches, unauthorized access, and manipulation due to centralized storage, outdated encryption protocols, and the lack of tamper- proof audit trails. Moreover, with the rise of quantum computing, traditional encryption standards like RSA and ECC are becoming obsolete. Existing systems also fail to integrate real- time analysis of medical data. This project addresses these limitations by designing a secure system that employs post-quantum cryptography, decentralized blockchain logging, and ML- driven insights — ensuring data integrity, confidentiality, and proactive decision support.

2.3 EXISTING SYSTEM

In the traditional healthcare system, patient records are typically stored in centralized databases or physical files. Doctors manually access these records during appointments, while administrative staff handle storage, transfers, and retrieval. Data is often transferred through emails or shared folders with basic access controls, making it vulnerable to unauthorized access. Encryption, if used, is often outdated or improperly implemented. Backup mechanisms may also be manual or not regularly updated.

Limitations of Existing System

- Lack of Data Security: Most healthcare systems store data in centralized databases, which makes them vulnerable to hacking, data breaches, and insider threats.

- No Tamper Evidence: Traditional systems cannot easily detect if medical records have been modified or deleted. This reduces trust because it is difficult to verify the integrity of patient data.

- Manual Access Control: User permissions are often managed manually, which makes it difficult to track who accessed the data. This may lead to unauthorized access or misuse of confidential information.

- No Scalability: Conventional databases may struggle to manage the large and growing amount of healthcare data, which can affect system performance and reliability.

2.4 PROPOSED SYSTEM

This project proposes a secure healthcare data management system that integrates blockchain technology and post-quantum cryptography to protect sensitive medical records. The system is developed using the Django framework as the backend and connects to the Ethereum blockchain through Web3 integration. In this architecture, when a patient uploads a medical report, the file is first encrypted using quantum- resistant algorithms such as Kyber for key exchange and Falcon for encryption. The encrypted file is stored securely, while its metadata is recorded on the blockchain to ensure immutability and transparency. Doctors can access the encrypted reports, generate prescriptions, and securely share them with patients. The blockchain layer records transactions such as registration, appointments, and prescriptions in a tamper- proof ledger, while the quantum encryption layer ensures that only authorized users can decrypt and view the data. Overall, the system provides secure communication, data integrity, privacy protection, and quantum-resistant security for future healthcare systems.

Advantages of the Proposed System

- Quantum-Safe Encryption: The system uses post-quantum algorithms like Kyber and Falcon to encrypt healthcare data. These algorithms protect patient records from both current cyber threats and future quantum computer attacks.

- Blockchain Integrity: Blockchain stores transaction details in a tamper- proof ledger, ensuring that medical records cannot be modified or deleted. This provides data integrity, transparency, and traceability.

- Decentralized Access: The system uses a decentralized blockchain network, which removes the risk of a single point of failure and improves security compared to traditional centralized systems.

- End-to-End Encryption: Healthcare data is encrypted during storage and transmission, ensuring that only the intended users can read the information and maintaining patient privacy.

2.5 OBJECTIVES

- Secure File Upload and Sharing Platform: The objective is to develop a secure platform for uploading and sharing healthcare data such as medical reports, prescriptions, and diagnostic results. This ensures that patient information can be stored and exchanged safely between doctors and patients without the risk of unauthorized access.

- Implementation of Post-Quantum Encryption: This project aims to implement post-quantum cryptographic algorithms such as FALCON and KYBER to protect healthcare files. KYBER is used for secure key exchange, while FALCON is used for encryption and digital signatures, ensuring strong protection against both current and future cyber threats.

- Blockchain-Based Transaction Logging: Another objective is to record all healthcare data transactions on a blockchain network. Blockchain provides an immutable and tamper-proof ledger, which helps in tracking data access, maintaining transparency, and ensuring secure access management.

- Integration of Machine Learning Models: The system integrates machine learning models such as Decision Tree Classifier (DTC) and Random Forest Classifier (RFC) to analyze decrypted healthcare data.

2.6 HARDWARE & SOFTWARE REQUIREMENTS

2.6.1 HARDWARE REQUIREMENTS

Hardware interfaces specifies the logical characteristics of each interface between the software product and the hardware components of the system. The following are some hardware requirements,

- Processor : Intel Core i5
- Hard disk : 512GB
- RAM : 8GB

2.6.2 SOFTWARE REQUIREMENTS

Software Requirements specifies the logical characteristics of each interface and software components of the system. The following are some software requirements,

- Operating system : Windows
- Language : Python
- Database : Redis
- Frame Work : Django

3.SYSTEM ARCHITECTUE & DESIGN

3. SYSTEM ARCHITECTURE & DESIGN

Project architecture refers to the structural framework and design of a project, encompassing its components, interactions, and overall organization. It provides a clear blueprint for development, ensuring efficiency, scalability, and alignment with project goals. Effective architecture guides the project's lifecycle, from planning to execution, enhancing collaboration and reducing complexity.

3.1 PROJECT ARCHITECTURE

This architecture shows a secure healthcare system where patients and doctors use a Django based web application to upload, view, and manage medical reports. The reports are encrypted using Kyber post-quantum encryption and transactions are recorded on the blockchain to ensure security and data integrity.

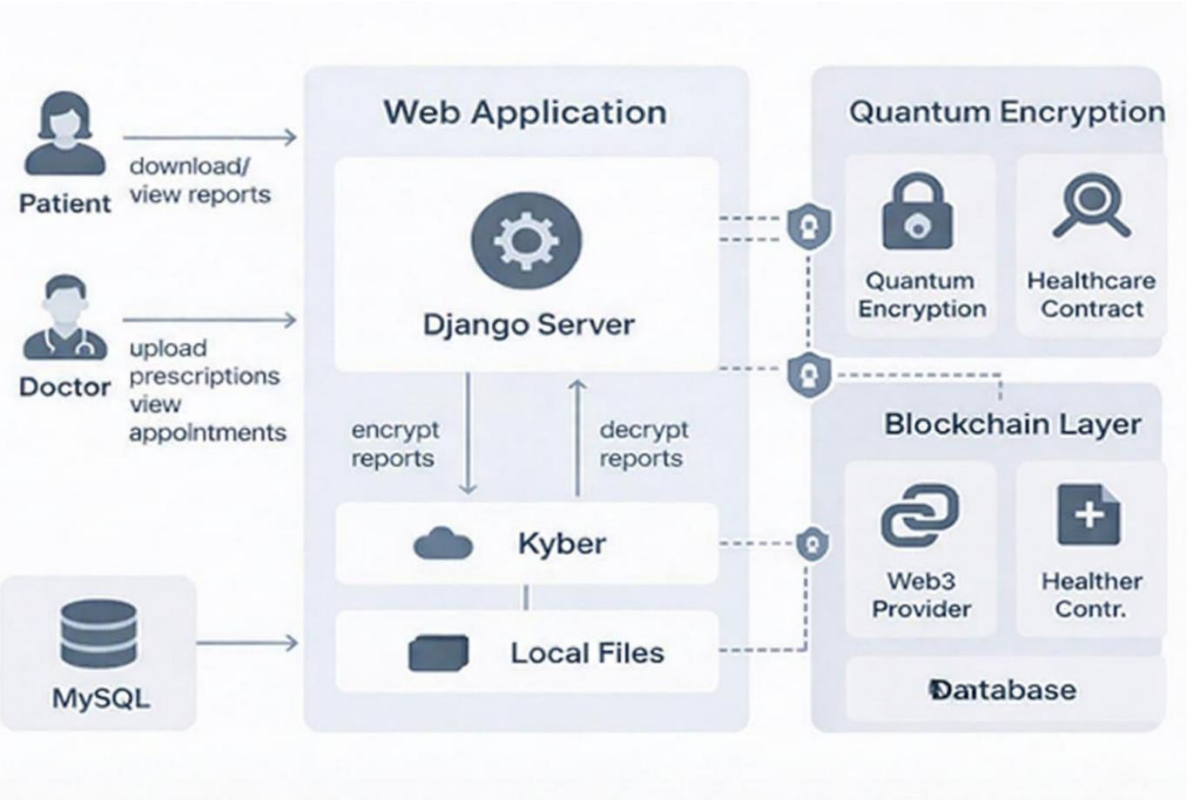

Figure 3.1: Project Architecture of QuantumHealthChain : A Post-Quantum Kyber Falcon Key Exchange and Blockchain-Backed Framework For Tamper proof of Electronic Health Records

3.2 DESCRIPTION

Secure Key Exchange: The system first establishes a secure communication channel between the patient and doctor using the CRYSTALS-Kyber post-quantum algorithm. Kyber securely exchanges encryption keys so that both sender and receiver can communicate safely without the risk of interception.

File Encryption: When a medical file such as a report or prescription is uploaded, it is encrypted using FALCON. This ensures that the healthcare data remains protected and cannot be accessed by unauthorized users.

Blockchain Storage: The encrypted file's metadata and transaction details are stored on the Ethereum blockchain using smart contracts. This creates a tamper-proof record that maintains data integrity and transparency.

Secure File Retrieval: When an authorized user requests the file, the system verifies the access and retrieves the encrypted data. The shared key generated through Kyber is used to decrypt the file securely.

Backend Processing and Prediction: After decryption, the backend system built with Django processes the data. The system can run analysis, classification, or generate prescriptions based on the medical information.

Secure Result Storage: The generated output, such as prescriptions or predictions, is again encrypted and recorded on the blockchain to maintain security and traceability.
Authorized Access: Finally, only the intended patient or doctor can decrypt and view the results, ensuring privacy, confidentiality, and secure healthcare data management.

3.3 UML DIAGRAMS

Unified Modeling Language (UML) is a standardized visual language used to model, design, and document the architecture of software systems. It provides a set of graphical notations to represent the structure and behavior of a system, making complex systems easier to understand and communicate among developers, stakeholders, and business analysts.

3.3.1 CLASS DIAGRAM

The Class Diagram defines the static structure of the system by showing its main classes, their attributes, methods, and relationships. For our secure image- hiding application, it illustrates components such as key management, encryption modules(ECC/ChaCha20), compression/decompression, and the GUI. This view helps developers understand how data and functionality are organized, making it easier to plan implementations and identify class responsibilities.

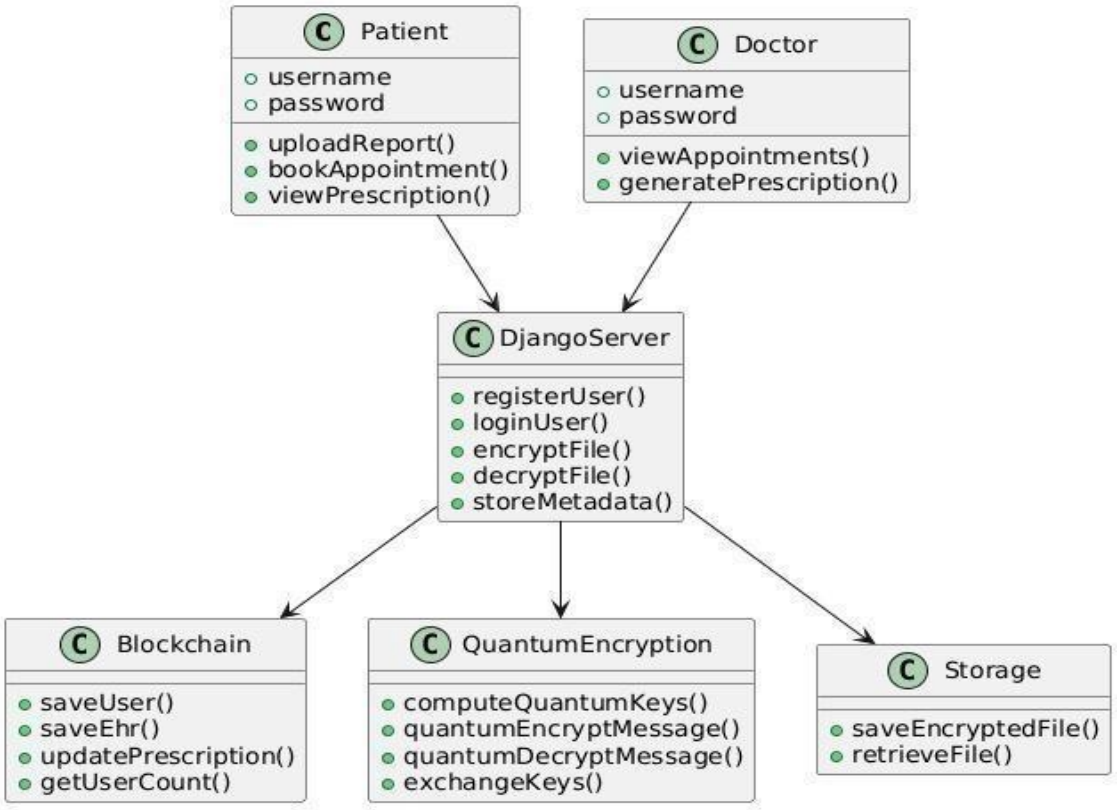

Figure 3.3.1: Class Diagram of QuantumHealthChain : A Post-Quantum Kyber Falcon Key Exchange and Blockchain-Backed Framework For Tamper proof of Electronic Health Records

3.3.2 USE CASE DIAGRAM

The Use Case Diagram identifies the actors (users or external systems) and their goals (use cases) within the system. In our application, two primary actors—Sender and Receiver—interact with use cases such as "Upload Image," "Encrypt & Embed," "Compress & Send," "Receive & Decompress," "Extract & Decrypt," and "Verify Integrity." This view clarifies user interactions and helps define system requirements from the end-user perspective.

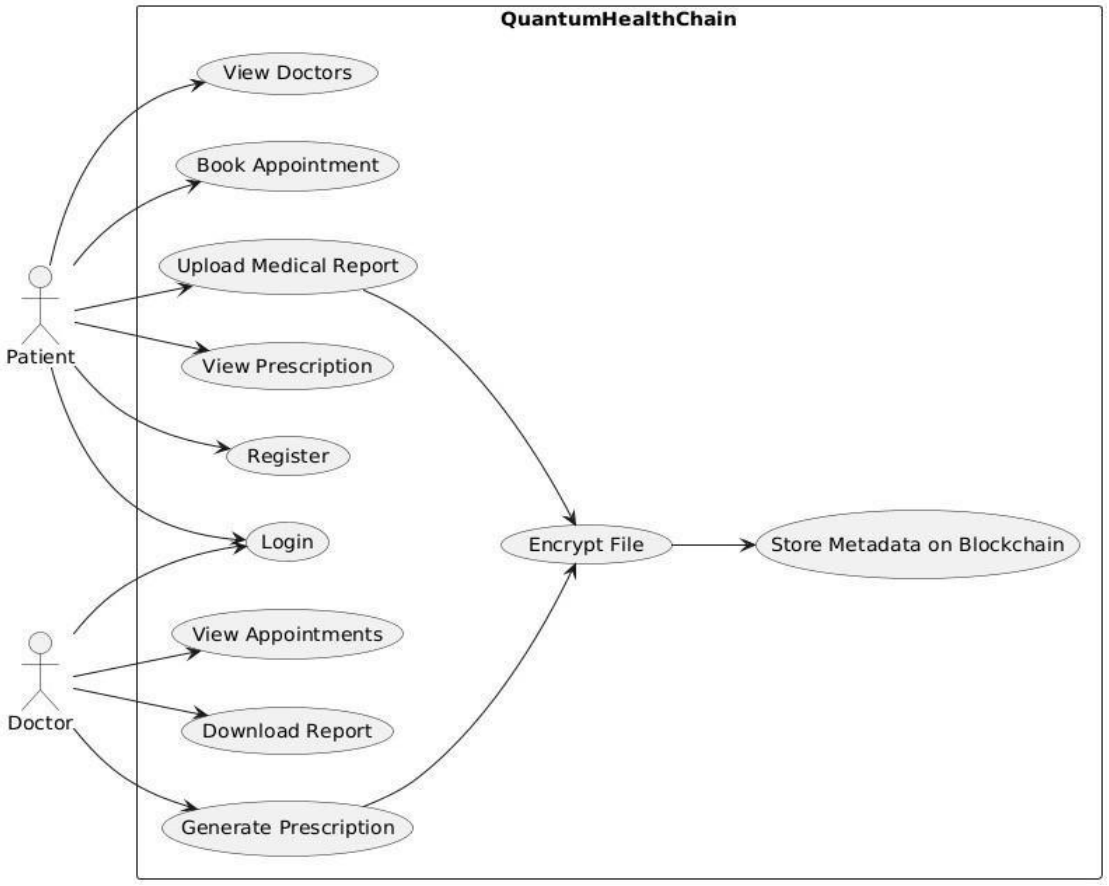

Figure 3.3.2: Use Case Diagram of QuantumHealthChain : A Post-Quantum Kyber Falcon Key Exchange and Blockchain-Backed Framework For Tamper proof of Electronic Health Records

3.3.3 SEQUENCE DIAGRAM

The Sequence Diagram illustrates the time-ordered interactions among objects during key scenarios: encrypting/embedding on the sender side and extracting/decrypting on the receiver side. It shows lifelines for the GUI, KeyManager, ECCModule/ChaChaModule, StegoEngine, Compressor, IntegrityChecker, and back to GUI. Messages such as getECCKeys(), encryptECC(), embedBits(), compressFile(), plus SHA computation and decryption calls appear in order. This view helps track the runtime flow of data and control between components.

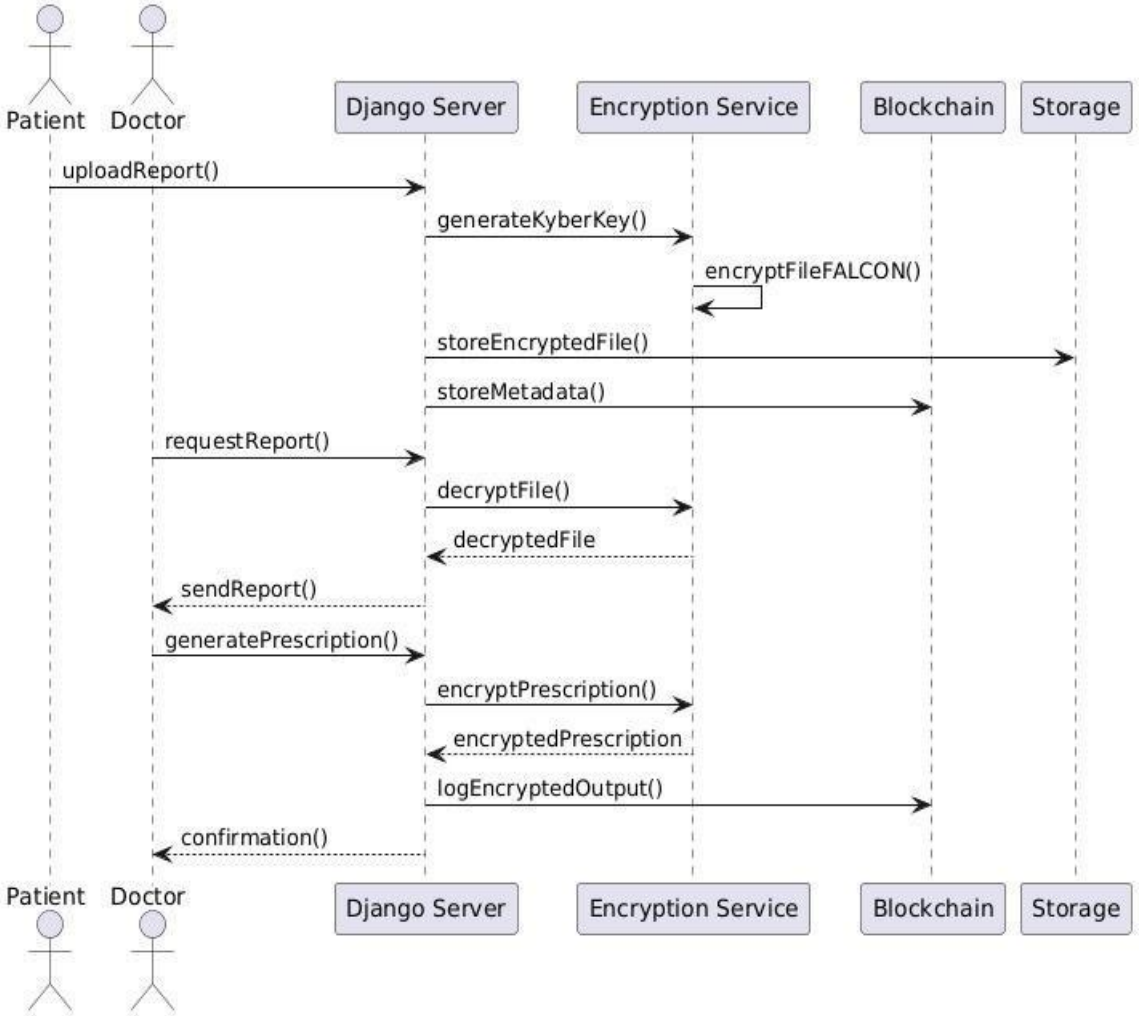

Figure 3.3.3: Sequence Diagram of QuantumHealthChain : A Post-Quantum Kyber Falcon Key Exchange and Blockchain-Backed Framework For Tamper proof of Electronic Health Records

3.3.4 ACTIVITY DIAGRAM

The Activity Diagram shows the workflow of the secure healthcare system for both patient and doctor. The patient logs in, views doctors, books an appointment, and uploads a medical report. The system generates a key using CRYSTALS-Kyber and encrypts the report using FALCON before storing it and logging details on the blockchain. The doctor verifies access, decrypts the report, generates a prescription, and the patient can later view or download it securely.

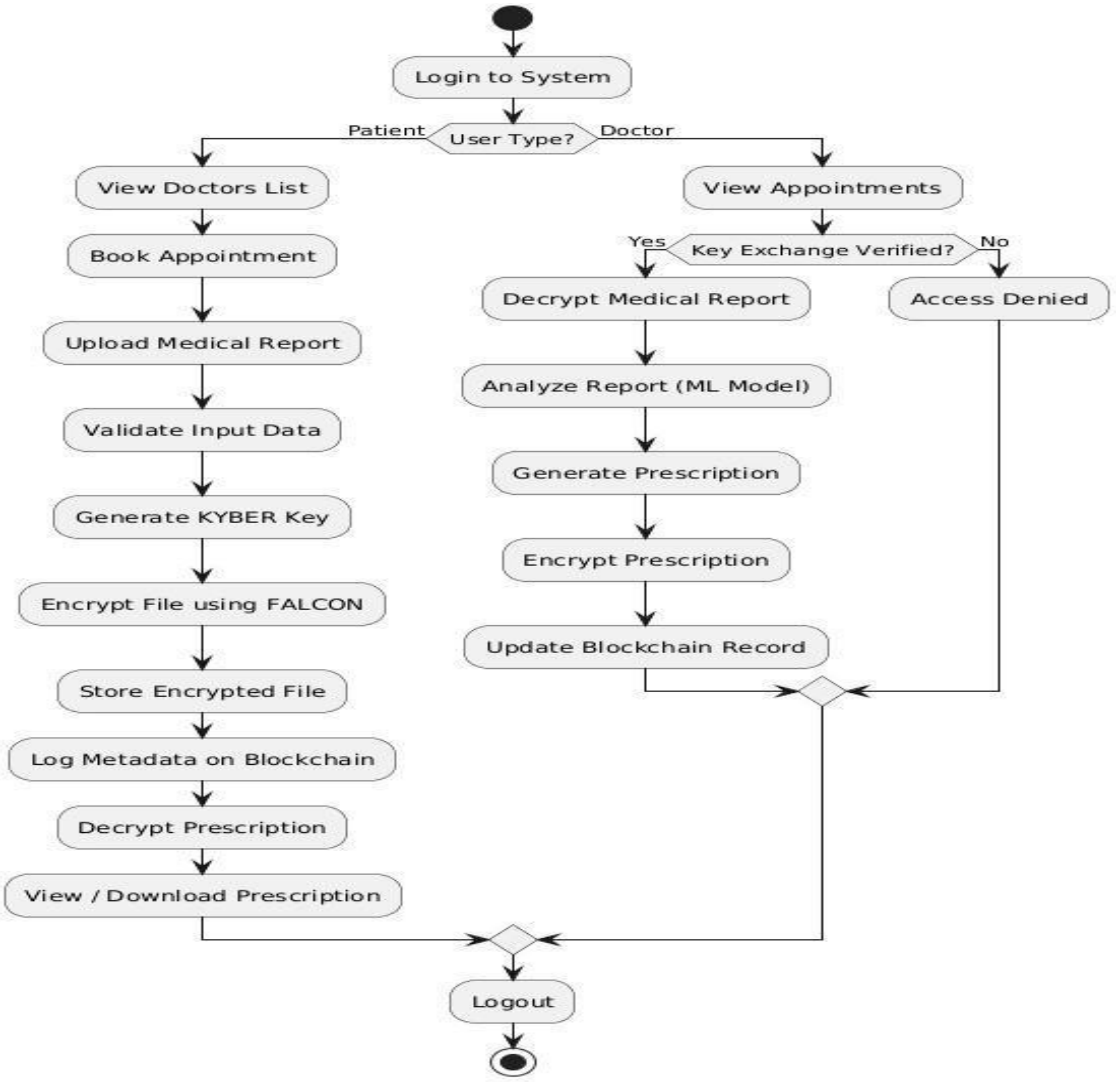

Figure 3.3.4: Activity Diagram of QuantumHealthChain : A Post-Quantum Kyber Falcon Key Exchange and Blockchain-Backed Framework For Tamper proof of Electronic Health Records

4. IMPLEMENTATION

4. IMPLEMENTATION

The implementation phase of a project involves executing the planned strategies and tasks. It requires meticulous coordination, resource allocation, and monitoring to ensure that objectives are met efficiently. Effective implementation is crucial for achieving project goals and delivering expected outcomes within the set timeline and budget constraints.

4.1 ALGORITHMS USED

Kyber Algorithm (Post-Quantum Key Exchange)

The project uses CRYSTALS-Kyber to securely exchange encryption keys between the patient and the doctor. It generates a shared secret key that is used to encrypt and decrypt healthcare files such as medical reports and prescriptions. Kyber is based on lattice-based cryptography and is designed to resist attacks from future quantum computers.

Advantages:

- Quantum-resistant security – Protects data from quantum computer attacks.
- Fast key generation – Efficient for secure communication systems.
- High security level – Based on hard mathematical lattice problems.
- Suitable for large-scale applications – Works well in cloud and healthcare systems.

Disadvantages:

- Larger key sizes compared to traditional algorithms.
- More computational resources may be required.
- Implementation complexity is higher than classical algorithms

FALCON Algorithm (Post-Quantum Digital Signature / Encryption)

The system uses FALCON to encrypt and digitally sign medical reports and prescriptions before storing or sharing them. It ensures that the files remain secure and cannot be altered by unauthorized users.

Advantages:

- Strong security against quantum attacks.
- Small signature size compared to many other post-quantum algorithms.
- High efficiency and fast verification.
- Ensures data integrity and authenticity.

Disadvantages:

- Complex mathematical implementation.
- Higher computational cost during signature generation.
- Requires careful implementation to avoid security risks.

The system is developed using the Django framework, which manages all user requests and application logic. It supports functions such as patient and doctor registration, login authentication, viewing doctors, booking appointments, uploading medical reports, generating prescriptions, and accessing encrypted files. Django connects the frontend interface with backend processing, allowing the system to handle data securely while interacting with the encryption module and blockchain network.

To ensure secure and tamper-proof data storage, the system integrates blockchain technology using Ethereum smart contracts. These smart contracts store important metadata such as user registration information, appointment records, and prescription updates. The backend communicates with the blockchain through Web3, enabling operations such as saving new users, recording electronic health record details, and updating prescriptions. Since blockchain data cannot be modified once recorded, it ensures transparency, integrity, and reliable tracking of healthcare records.

Sensitive healthcare data is protected using post-quantum cryptography. The system uses CRYSTALS-Kyber to perform secure key exchange between patients and doctors, while FALCON is used to encrypt medical reports and prescriptions before storage or transmission. When a file is uploaded, it is encrypted using a generated key and stored securely in the system. When an authorized user accesses the file, the system decrypts it using the same key after verifying the user.

All medical reports and prescription files are stored in encrypted form in the system directory. Before saving a file, the system checks whether the file already exists and replaces it if necessary. Only file metadata such as the filename, patient name, doctor name, and date are stored on the blockchain, which improves system efficiency while maintaining data security.

Patients can book appointments by selecting a doctor and uploading their medical reports. The system encrypts the uploaded file and stores it locally, while the appointment details and file reference are recorded on the blockchain. Doctors can log in to their dashboard to view assigned appointments, access encrypted reports, and generate prescriptions. Once a prescription is created, it is also encrypted and securely stored so that the patient can download it later.

The system also includes several security mechanisms such as role-based access control, encrypted file storage, and blockchain-based record management. Only authorized users are allowed to access sensitive healthcare data, and file decryption is performed only after successful key verification between the doctor and patient. These mechanisms ensure privacy, data integrity, and secure management of healthcare information.

4.2 SAMPLE CODE

```
from django.shortcuts import render from datetime import datetime

from django.template import RequestContext from django.contrib import messages

from django.http import HttpResponse from django.core.files.storage import

FileSystemStorage import os

from datetime import datetime import json

from web3 import Web3, HTTPProvider import pickle

import time import base64

import QuantumEncryption from QuantumEncryption import * import Kyber from Kyber

import *global username, doctor, prescriptionList, usersList global contract, web3

#function to call contract def getContract():

global contract, web3

blockchain_address = 'http://127.0.0.1:9545'

web3 = Web3(HTTPProvider(blockchain_address)) web3.eth.defaultAccount =

web3.eth.accounts[0] compiled_contract_path = 'Healthcare.json' #Healthcare contract file

deployed_contract_address='0xd374Cb05bd6187D6cF905D7bBD85f2b704fBDD29' #contract

address with open(compiled_contract_path) as file:

contract_json = json.load(file)  # load contract info as JSON

  contract_abi = contract_json['abi']  # fetch contract's abi - necessary to call its functions

file.close()

    contract = web3.eth.contract(address=deployed_contract_address, abi=contract_abi)

getContract()

def getUsersList():
```

```
global usersList, contract usersList = []

count = contract.functions.getUserCount().call() for i in range(0, count):

user =contract.functions.getUsername(i).call()

password=contract.functions.getPassword(i).c all() phone =

contract.functions.getPhone(i).call()

 email = contract.functions.getEmail(i).call() address = contract.functions.getAddress(i).call()

desc =contract.functions.getDescription(i).call() utype =
contract.functions.getUserType(i).call()

usersList.append([user, password, phone, email, address, desc, utype])

def getPrescriptionList(): global prescriptionList, contract prescriptionList = []

count = contract.functions.getEhrCount().call() for i in range(0, count):

uname = contract.functions.getPatient(i).call() docname =

contract.functions.getDoctor(i).call() disease= contract.functions.getDisease(i).call()

report = contract.functions.getReport(i).call()

prescription = contract.functions.getPrescription(i).call() ehr_date =

contract.functions.getEhrDate(i).call()

        prescriptionList.append([uname, docname, disease, report, prescription, ehr_date])

getUsersList()

getPrescriptionList()

def ViewReport(request): if request.method =='GET': global dec_time

 filename = request.GET.get('file', False) data =

quantumDecryptMessage("HealthcareApp/static/files/"+filename

response = HttpResponse(data,content_type='application/force- download') response['Content-

Disposition'] = 'attachment; filename='+filename

return response
```

```
def
GeneratePrescriptionAction(requ est): if request.method == 'POST':
global prescriptionList
pid = request.POST.get('t1', False) prescription = request.POST.get('t2',

False) filename = request.FILES['t3'].name myfile = request.FILES['t3'].read()
if os.path.exists("HealthcareApp/static/files/"+filename):
os.remove("HealthcareApp/static/files/"+filename)
key = computeQuantumKeys(myfile)
        iv, ciphertext = quantumEncryptMessage(myfile, key,
"HealthcareApp/static/files/"+filename)
contract.functions.updatePrescription(int(pid), prescription+"#"+filename).transact() pl =
prescriptionList[int(pid)] pl[4] = prescription+"#"+filename
data = "Prescription Updated Successfully" context= {'data':data}
return render(request,'DoctorScreen.html', context)

def
GeneratePrescription(requ est): if request.method == 'GET':
global username
pid = request.GET['pid']
        output = '<tr><td><font size="3" color="black">Appointment ID</td><td><input
type="text" name="t1" size="25" value="'+pid+'" readonly/></td></tr>'
context= {'data':output}
return render(request,'GeneratePrescription.html', context)
def
ViewAppointments(requ est): if request.method == 'GET':
```

```
global username, prescriptionList
output = '<table border=1 align=center>'
output+='<tr><th><font size=3 color=black>Appointment ID</font></th>'
output+='<th><font size=3 color=black>Patient Name</font></th>' output+='<th><font
size=3 color=black>Doctor Name</font></th>' output+='<th><font size=3
color=black>Disease Details</font></th>' output+='<th><font size=3 color=black>Report
Name</font></th>' output+='<th><font size=3 color=black>Prescription</font></th>'
output+='<th><font size=3 color=black>Booking Date</font></th>' output+='<th><font
size=3 color=black>View Report</font></th>' output+='<th><font size=3
color=black>Generate Pescription</font></th></tr>'
for i in
range(len(prescription List)): pl = prescriptionList[i]
if pl[1] == username and exchangeKeys(pl[0], pl[1]): output+='<tr><td><font size=3
color=black>'+str(i+1)+'</font></td>' output+='<td><font size=3
color=black>'+str(pl[0])+'</font></td>'output+='<td><fontsize=3
color=black>'+str(pl[1])+'</font></td>'output+='<td><fontsize=3
color=black>'+str(pl[2])+'</font></td>'output+='<td><fontsize=3
color=black>'+str(pl[3])+'</font></td>'output+='<td><fontsize=3
color=black>'+str(pl[4])+'</font></td>'output+='<td><fontsize=3
color=black>'+str(pl[5])+'</font></td>'
  output+='<td> <a href=\'ViewReport?file='+pl[3]+'\'> <font size=3color=blue>Click
Here</font></a></td>'
if pl[4] == 'None':
                output+='<td><ahref=\'GeneratePrescription?pid='+str(i)+'\'><fontsize=3
color=blue>Click Here for Prescription</font></a></td></tr>'
```

```
else:
        output+='<td><font size=3 color=black>'+pl[4]+'</font></td></tr>' context=

{'data':output} return render(request,'DoctorScreen.html', context)

def

ViewPrescription(req uest): if request.method == 'GET':

global prescriptionList, username

output = '<table border=1 align=center>'

output+='<tr><th><font size=3 color=black>Patient Name</font></th>' output+='<th><font

size=3 color=black>Doctor Name</font></th>' output+='<th><font size=3

color=black>Disease Details</font></th>' output+='<th><font size=3 color=black>Report

Name</font></th>' output+='<th><font size=3 color=black>Prescription</font></th>'

output+='<th><font size=3 color=black>Prescription File Name</font></th>'

output+='<th><font size=3 color=black>Download Prescription</font></th>'

output+='<th><font size=3 color=black>Date</font></th></tr>'

for i in range(len(prescriptionList)): pl = prescriptionList[i]

if pl[0] == username: prescription = pl[4] print(prescript ion) pres = "None" fname

arr = "-"file_hash = "- "

if prescription != 'None':

prescription.split("#") pres = arr[0]

fname = arr[1]

output+='<tr><td><font size=3 color=black>'+pl[0]+'</font></td>' output+='<td><font size=3

color=black>'+pl[1]+'</font></td>' output+='<td><font

size=3 color=black>'+pl[2]+'</font></td>' output+='<td><font size=3
```

```
color=black>'+pl[3]+'</font></td>' output+='<td><font size=3
color=black>'+pres+'</font></td>' output+='<td><font size=3
color=black>'+fname+'</font></td>' if prescription == 'None':
    output+='<td><font size=3 color=black>Pending</font></td>' else:
output+='<td><a href=\'ViewReport?file='+fname+'\'><font
size=3 color=blue>Click Here</font></a></td>'
output+='<td><font size=3 color=black>'+pl[5]+'</font></td></tr>' output +=
"</table><br/><br/><br/><br/>"
context= {'data':output}
return render(request,'PatientScreen.html', context)
def
AppointmentAction(request):
if request.method == 'POST':
global username
doctor = request.POST.get('t1', False) patient = request.POST.get('t2', False) disease =
request.POST.get('t3', False) today = str(datetime.now()) filename = request.FILES['t4'].name
myfile = request.FILES['t4'].read()
if os.path.exists("HealthcareApp/static/files/"+filename):
os.remove("HealthcareApp/static/files/"+filename)
key = computeQuantumKeys(myfile)
        iv, ciphertext = quantumEncryptMessage(myfile, key,
"HealthcareApp/static/files/"+filename)
        msg = contract.functions.saveEhr(patient, doctor, disease, filename, 'None',
today).transact()
```

```
tx_receipt = web3.eth.waitForTransactionReceipt(msg)

prescriptionList.append([patient, doctor, disease, filename, 'None', today])

        context= {'data':'Your Appointment Confirmed with Doctor :
'+doctor+'<br/>Appointment ID : '+str(len(prescriptionList))+'<br/>'+str(tx_receipt)}

        return render(request, 'PatientScreen.html', context) def Appointment(request):

if request.method ==GET': global username doctor = request.GET['doctor'] today =

datetime.now() month =

today.month year = today.year  day= today.day

        output = '<tr><td><font size="3" color="black">Doctor Name</td><td><input

type="text" name="t1" size="25" value="'+doctor+'" readonly/></td></tr>'

output  += '<tr><td><font  size="3"
color="black">Patient Name</td><td><input type="text" name="t2" size="25"
value="'+username+'" readonly/></td></tr>'

        output += '<tr><td><font size="3"
color="black">Disease Details</td><td><input type="text" name="t3" size="40"
/></td></tr>'

        output += '<tr><td><font size="3"
color="black">Upload Medical Report</td><td><input type="file" name="t4"
size="40" /></td></tr>'

context= {'data':output}

 return render(request,'BookAppointment.html', context) def

BookAppointment(request

): if request.method ==global usersList

output = '<table border=1 align=center>'

output+='<tr><th><font size=3 color=black>Doctor Name</font></th>' output+='<th><font

size=3 color=black>Phone No</font></th>' output+='<th><font size=3 color=black>Email

ID</font></th>' output+='<th><font size=3 color=black>Address</font></th>'
```

```
output+='<th><font size=3 color=black>Description</font></th>' output+='<th><font size=3
color=black>Book Appointment</font></th></tr>' for i in range(len(usersList)):
ul = usersList[i]
if ul[6] == 'Doctor'
output+='<tr><td><font size=3 color=black>'+ul[0]+'</font></td>' output+='<td><font
size=3 color=black>'+ul[2]+'</font></td>' output+='<td><font size=3
color=black>'+ul[3]+'</font></td>' output+='<td><font size=3
color=black>'+ul[4]+'</font></td>' output+='<td><font size=3
color=black>'+ul[5]+'</font></td>'
output+='<td><a href=\'Appointment?doctor='+ul[0]+'\'><font size=3
color=blue>Click Here to Book Appointment</font></a></td></tr>'
output += "</table><br/><br/><br/><br/>" context= {'data':output}
return render(request,'PatientScreen.html', context)
def index(request):
if request.method == 'GET':
return render(request,'index.html', {})
def Register(request):
if request.method == 'GET':
return render(request, 'Register.html', {})
def DoctorLogin(request):
if request.method == 'GET':
    return render(request, 'DoctorLogin.html', {}) def PatientLogin(request):
if request.method == 'GET':
return render(request, 'PatientLogin.html',
```

```
{}) def isUserExists(username): is_user_exists = False
    global details
        mysqlConnect = pymysql.connect(host='127.0.0.1',port = 3306,user = 'root', password =
    'root', database = 'doctorpatientapp',charset='utf8')
    with mysqlConnect:
    result = mysqlConnect.cursor() result.execute("select * from user_signup where
    username='"+username+"'") lists = result.fetchall() for ls in lists:
    is_user_exists =
    True
    return is_user_exists
    def RegisterAction(request): if request.method ==
    'POST': global usersList  username = request.POST.get('t1', False) password =
    request.POST.get('t2', False) contact = request.POST.get('t3', False) email =
    request.POST.get('t4', False)
    address = request.POST.get('t5', False) desc = request.POST.get('t6', False) usertype =
    request.POST.get('t7', False)
    count = contract.functions.getUserCount().call() status = "none"
    for i in range(0, count): user1 = contract.functions.getUsername(i).call()
    if  username == user1
    status = "exists" break
    if status == "none":
            msg = contract.functions.saveUser(username, password, contact, email, address, desc,
    usertype).transact()
    tx_receipt = web3.eth.waitForTransactionReceipt(msg) usersList.append([username,
    password, contact, email, address, desc, usertype]) context= {'data':'Signup Process
```

```
Completed<br/>'+str(tx_receipt)} return render(request, 'Register.html',
context) else:
context= {'data':'Given username already exists'}
 return render(request, 'Register.html', context) def PatientLoginAction(request):
if request.method == 'POST':
global username, contract, usersList username  =  request.POST.get('t1', False)
password= request.POST.get('t2', False)
status= 'none'
for i in
range(len(usersList)
): ulist = usersList[i]

user1 = ulist[0] pass1 = ulist[1] utype = ulist[6]
if user1 == username and pass1 == password and utype == "Patient": status = "success"
break
if status == 'success': output = 'Welcome '+username context=
{'data':output}
    return render(request, "PatientScreen.html", context) if status == 'none':
context= {'data':'Invalid login details'}
return render(request, 'PatientLogin.html', context)
def
DoctorLoginAction(requ est): if request.method
== 'POST'
global username, contract, usersList username = request.POST.get('t1', False)

Password = request.POST.get('t2', False) status = 'none'
```

```
for i in

range(len(usersList)

): ulist = usersList[i] user1 =

ulist[0] pass1 = ulist[1] utype = ulist[6]

if user1 == username and pass1 == password and utype == "Doctor": status = "success"

break

if status == 'success':

output = 'Welcome '+username context= {'data':output}

return render(request, "DoctorScreen.html",

context) if status == 'none':

context= {'data':'Invalid login details'}

return render(request, 'DoctorLogin.html', context)
```

5.RESULT & DISCUSSION

5. RESULTS & DISCUSSION

The following screenshots showcase the results of our project, highlighting key features and functionalities. These visual representations provide a clear overview of how the system performs under various conditions, demonstrating its effectiveness and user interface. The screenshots serve as a visual aid to support the project's technical and operational achievements.

5.1 Home page:

Figure 5.1 shows the main homepage of the "Quantum Resistance Blockchain for Secure Health Data" system, which acts as the entry point for users. The page provides navigation options such as Patient Login, Doctor Login, and New User Signup, allowing different users to access the system securely. The interface visually represents the healthcare and blockchain concept, giving users a simple way to log in or register.

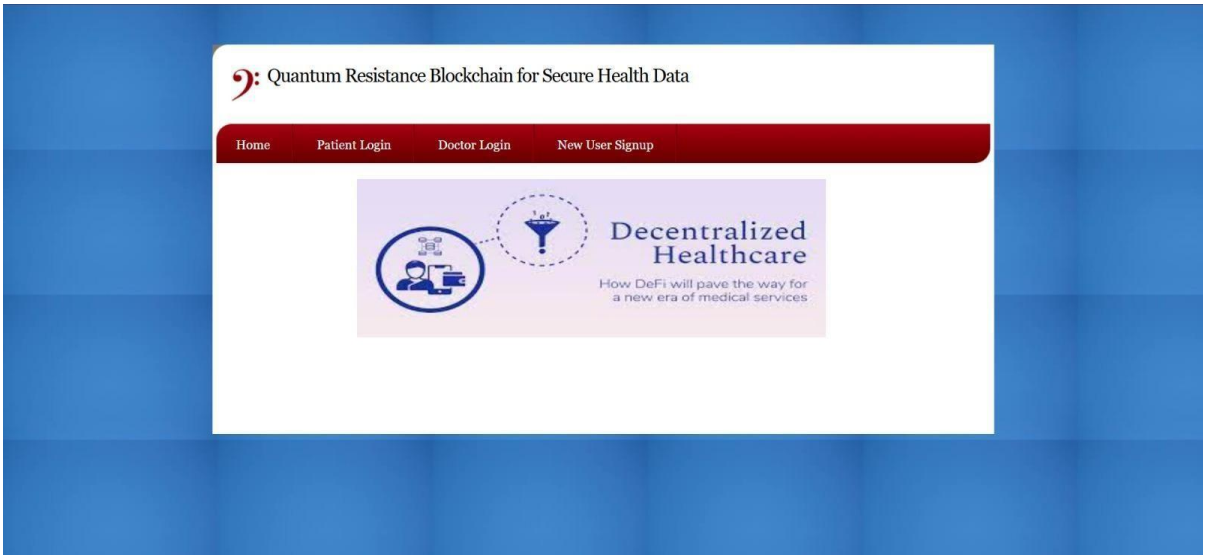

Figure 5.1 : Home page for Quantum resistance blockchain for secure health data.

5.2 Patient Signup Image :

Figure 5.2 shows the confirmation page for the New User Signup process in the “Quantum Resistance Blockchain for Secure Health Data” system. The page displays “Signup Process Completed” along with blockchain transaction details such as transaction hash and block information, confirming that the user registration has been successfully recorded on the blockchain. It also provides login fields so the user can log in to the system.

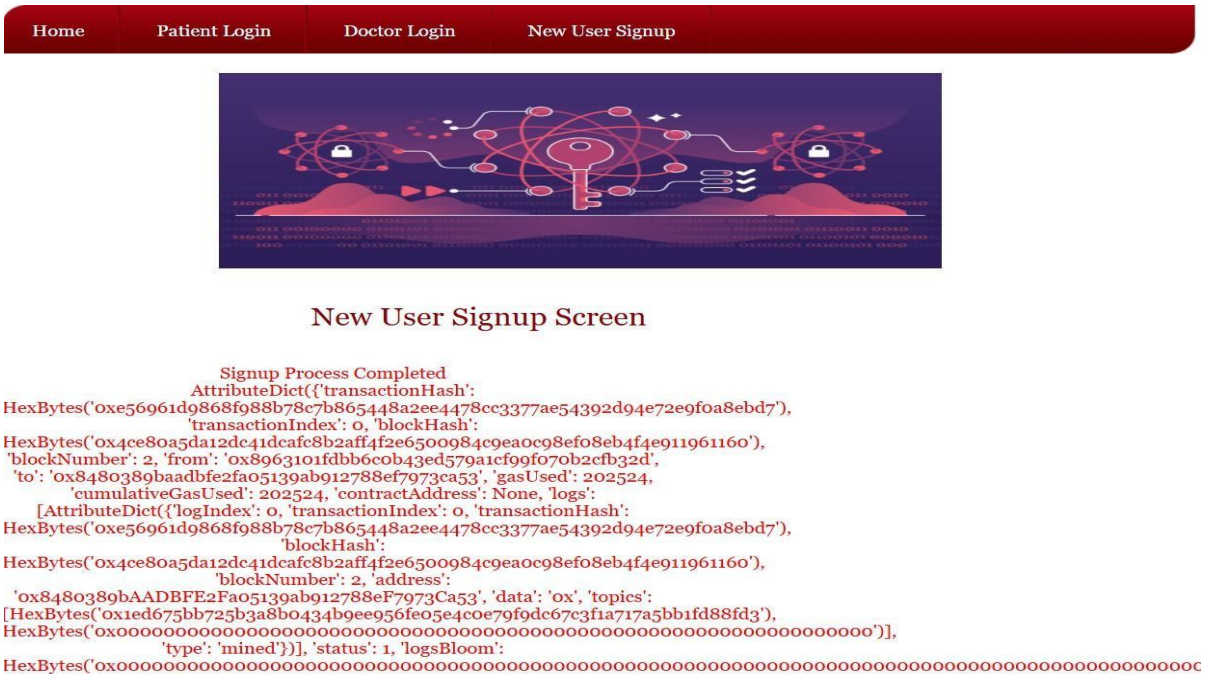

Figure 5.2 : Patient Signup for the Quantum resistance blockchain for secure health data.

5.3 Doctor Signup Image :

The Figure 5.3 Displays the successful registration confirmation for a doctor on the "Quantum Resistance Blockchain for Secure Health Data" platform. After a doctor completes the signup process on the "New User Signup Screen," this page appears, confirming that the "Signup Process Completed" successfully. The central feature is the detailed blockchain transaction receipt, which includes data like the 'transactionHash' and 'blockHash'. This serves as transparent, immutable proof that the new doctor's account has been securely created and recorded on the blockchain.

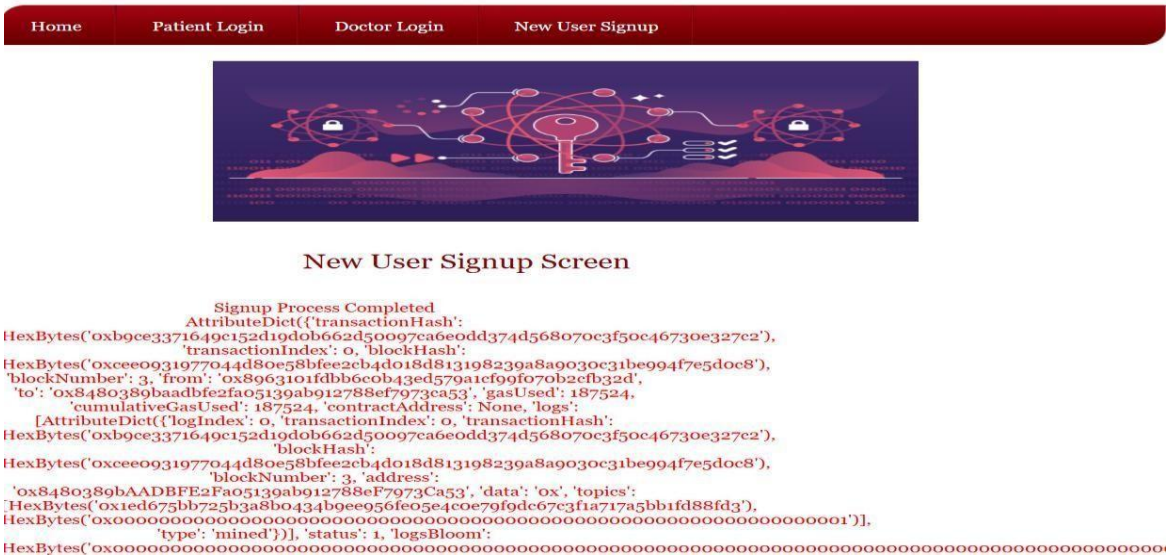

Figure 5.3: Doctor Signup for the Quantum resistance blockchain for secure health data.

5.4 Doctors List in Patient's Page :

Figure 5.4 shows the "View Doctors List" page from the patient dashboard in the "Quantum Resistance Blockchain for Secure Health Data" system. The page displays a table with doctor details such as name, phone number, email, address, and description. Patients can easily select a doctor and click "Book Appointment" to schedule a consultation.

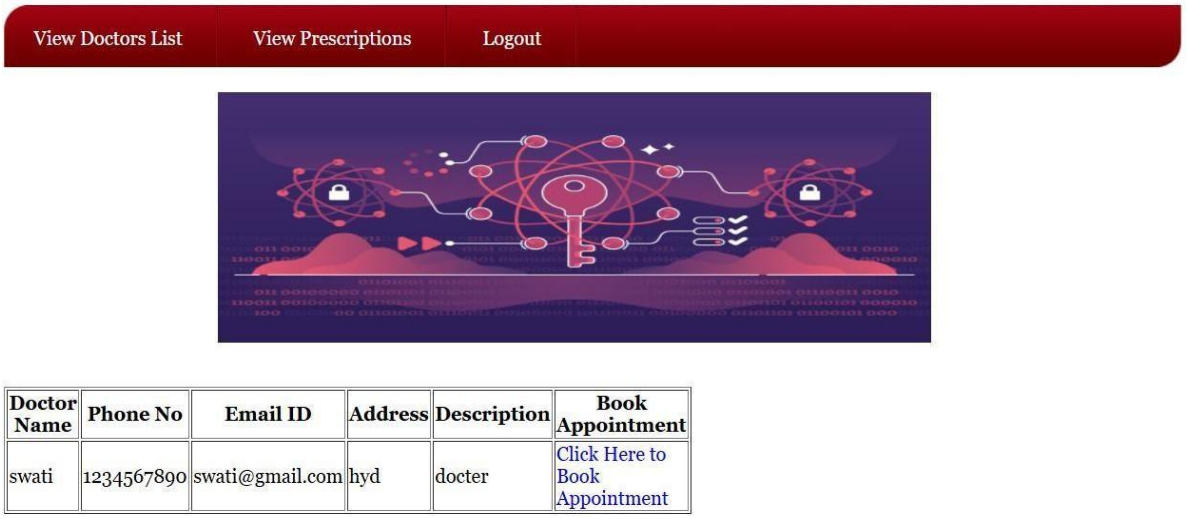

Figure 5.4 : View Doctors List in patient's page for the Book appointment.

5.5 Patient Confirmed Appointment :

Figure 5.5 shows the appointment confirmation page in the "Quantum Resistance Blockchain for Secure Health Data" system. It confirms that the patient's appointment with the doctor has been successfully booked and displays a unique Appointment ID. The page also shows blockchain transaction details, proving that the appointment information has been securely recorded on the blockchain.

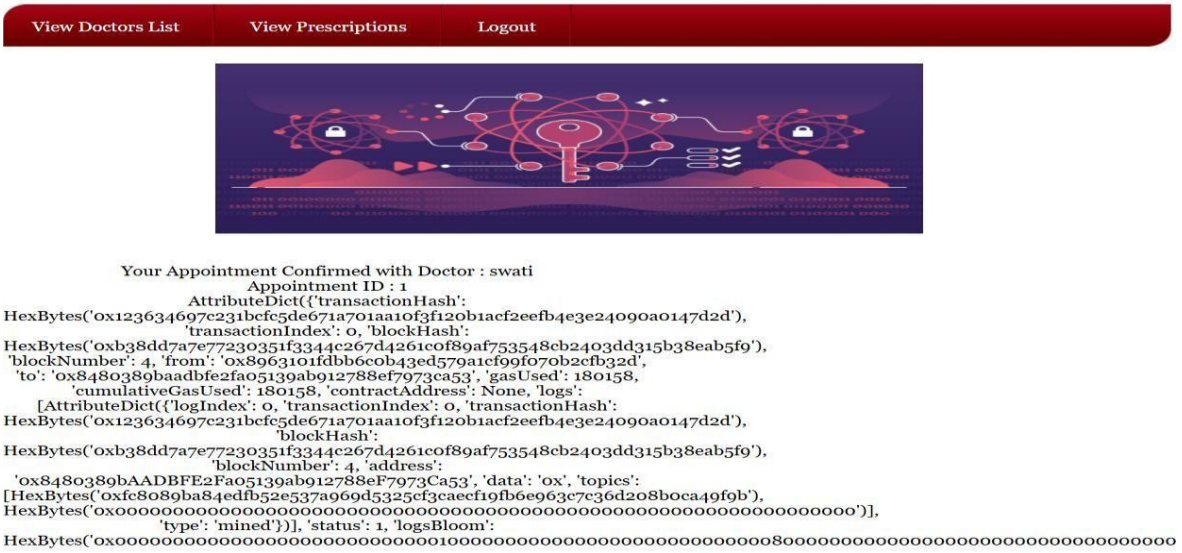

Figure 5.5: Patient Confirmed Appointment with doctor.

5.6 Doctor Dashboard:

Figure 5.6 shows the confirmation page in the doctor's dashboard of the "Quantum Resistance Blockchain for Secure Health Data" system. It displays the message "Prescription Updated Successfully," indicating that the doctor has successfully submitted or updated the patient's prescription. This confirms that the prescription data has been securely recorded on the blockchain.

Figure 5.6: Doctor Dashboard with prescription upload successfully.

5.7 Patient Dashboard:

Figure 5.7 shows the "View Appointments" page in the doctor's dashboard of the "Quantum Resistance Blockchain for Secure Health Data" system. The page displays a table containing patient details, disease information, prescription, and appointment date. It also provides options to view reports and generate prescriptions for each appointment.

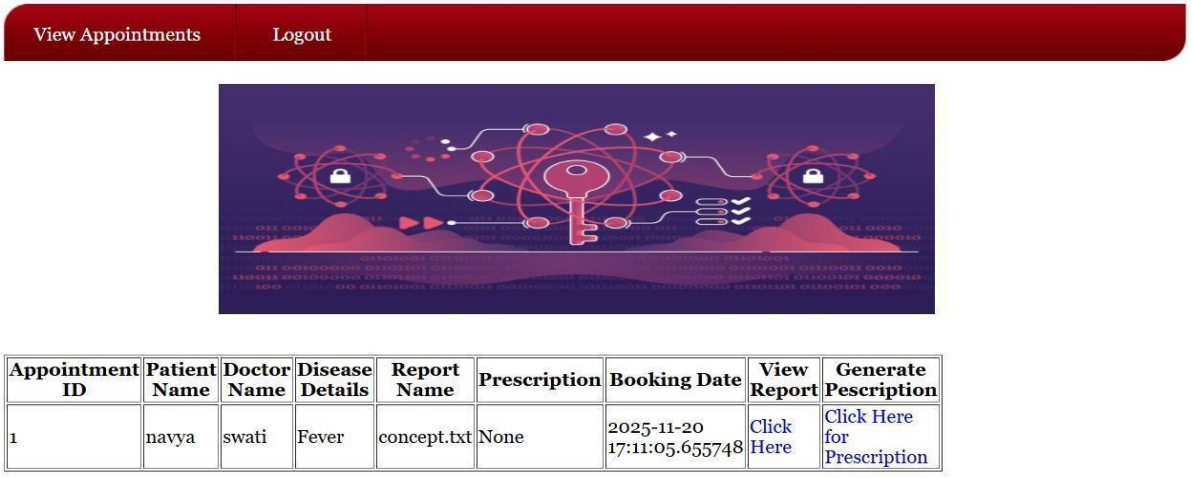

Figure 5.7 : Doctors prescription shows in patient dashboard.

6.VALIDATION

6. VALIDATION

The validation of this project mainly depends on systematic testing and verification to ensure the security, reliability, and proper functioning of the proposed healthcare system. The testing process focuses on verifying the quantum-safe encryption, blockchain transactions, secure file handling, and doctor–patient communication workflow. By implementing structured validation methods, the system ensures that sensitive healthcare data is securely stored, transmitted, and accessed only by authorized users while maintaining data integrity and privacy.

6.1 INTRODUCTION

Initially, the system is tested by validating user registration, login authentication, and role- based access control for both patients and doctors. Different test cases are executed to ensure that only registered users can access the system and perform operations such as booking appointments, uploading medical reports, and generating prescriptions.

Further validation is performed on the post-quantum cryptographic modules, where uploaded medical reports and prescriptions are encrypted using CRYSTALS- Kyber for secure key exchange and FALCON for encryption and digital signatures. The system verifies that files are properly encrypted before storage and correctly decrypted only by authorized users.

The project also validates the blockchain integration by testing smart contract transactions on Ethereum. Each operation such as user registration, appointment booking, and prescription updates is recorded as a blockchain transaction. The generated transaction hashes and block numbers confirm that the data is stored in a tamper-proof and immutable ledger.Finally, the system undergoes end-to-end workflow testing, where real scenarios such as patient appointment booking, doctor report access, prescription generation, and secure file retrieval are executed. This testing ensures that the entire system operates smoothly while maintaining high levels of data security.

6.2 TEST CASES

TABLE 6.2.1 USER REGISTRATION

Test case ID	Test case name	Purpose	Test Case	Output
1	User Registration	To register a new user in the system	User enters username,passw-ord,contact,email and user type	User successfully registered and stored on blockchain

TABLE 6.2.2 USER LOGIN

Test case ID	Test case name	Purpose	Input	Output
1	Patient Login	To verify patient login	Valid patient username and password	Patient dashboard opened
2	Doctor Login	To verify doctor login	Valid doctor username and password	Doctor dashboard opened

TABLE 6.2.3 APPOINTMENT BOOKING

Test case ID	Test case name	Purpose	Test Case	Output
1	Book Appointment	To schedule consultation with doctor	Patient selects doctor and uploads medical report	Appointment successfully confirmed

TABLE 6.2.4 PRESCRIPTION GENERATION

Test case ID	Test case name	Purpose	Test Case	Output
1	Generate Prescription	To create prescription for patient	Doctor enters prescription and uploads file	Prescription updated successfully

TABLE 6.2.5 VIEW REPORT

Test case ID	Test case name	Purpose	Test Case	Output
1	View Medical Report	To access encrypted report	Doctor clicks view report option	Report decrypted and downloaded securely

7.CONCLUSION & FUTURE ASPECTS

7. CONCLUSION & FUTURE ASPECTS

In conclusion, the project has successfully achieved its objectives, showcasing significant progress and outcomes. The implementation and execution phases were meticulously planned and executed, leading to substantial improvements and insights. Looking ahead, the future aspects of the project hold immense potential. Future developments will focus on expanding the scope, integrating new technologies, and enhancing sustainability. These advancements will not only strengthen the existing framework but also open new avenues for growth and innovation, ensuring the project remains relevant and impactful in the long term. This strategic approach will drive continuous improvement and success.

7.1 PROJECT CONCLUSION

The integration of quantum cryptography with blockchain technology in the proposed healthcare system significantly enhances the security and reliability of electronic health records. Traditional cryptographic algorithms such as RSA and AES, though robust against classical threats, are vulnerable to quantum computing attacks. This system mitigates those vulnerabilities by employing post-quantum algorithms like Kyber for secure key exchange and Falcon for data encryption and decryption. These algorithms are resistant to known quantum threats, ensuring that even with the rise of quantum computing, patient data remains secure. Additionally, the use of blockchain ensures data immutability and transparency, eliminating the risks associated with centralized data storage and unauthorized tampering by administrators. Smart contracts manage user interactions and data transactions securely and autonomously, further enhancing trust in the system. The combination of these advanced technologies demonstrates a practical, scalable, and highly secure solution for modern healthcare data management.

7.2 FUTURE ASPECTS

As quantum computing continues to evolve, the proposed system can be extended to support more advanced post-quantum cryptographic protocols to stay ahead of emerging threats. The current framework can be scaled to integrate with national healthcare databases, enabling secure inter-hospital data sharing and global interoperability without compromising patient privacy. Future enhancements may include the use of machine learning models to analyze encrypted health data for predictive diagnostics while maintaining confidentiality using homomorphic encryption. Furthermore, the blockchain component can be integrated with other decentralized systems such as decentralized identity management (DID) to give patients full control over their personal health data. The system can also benefit from real-time monitoring and alerts using Internet of Things (IoT) devices, securely linked through the blockchain network. Overall, this framework lays a strong foundation for a next- generation, tamper-proof, and quantum-resilient healthcare ecosystem.

8. BIBLIOGRAPHY

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8.2 GITHUB LINK

https://github.com/BatchuNikhil/Batch-17-Quantum-KYBER-FALCON-Key-Exchange-and-Blockchain-For-Securing-Electronic-Health-Records